THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL UNIT TEST SET 1

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A 14-year-old girl is shorter than her 14-year-old male cousin. The best explanation is: 1

(a) she is ill
(b) she is not eating
(c) growth rates vary and the cousin may have started his spurt
(d) she is old

2. Two rods A and B of equal size are heated at one end. A's pins fall in 20 s, B's pins fall in 60 s. This shows that: 1

(a) A is a better conductor **(b)** B is a better conductor
(c) both equal **(d)** neither conducts

3. A train 150 m long crosses a platform 250 m long in 20 s. The speed of the train is: 1

(a) 8 m/s **(b)** 12.5 m/s **(c)** 20 m/s **(d)** 7.5 m/s

4. A person has a blocked bile duct. The food whose digestion will be most affected is: 1

(a) water **(b)** fat-rich food
(c) starch-rich food **(d)** protein-rich food

5. A boy's voice is 'cracking'. This is temporary because: 1

(a) he is cold **(b)** he is ill
(c) larynx is growing and adjusting **(d)** voice is lost

6. At a seaside at noon, the sea breeze blows. At 11 p.m. the breeze direction is: 1

(a) reversed **(b)** same **(c)** upward **(d)** zero always

7. Two cyclists A and B start from the same point. A goes at 15 km/h and B at 20 km/h. After 3 h, B is ahead of A by: 1

(a) 15 km **(b)** 35 km **(c)** 60 km **(d)** 5 km

8. A man's heart rate is 80 beats per minute. In 24 hours, it beats: 1

(a) 80 times **(b)** 1,15,200 times **(c)** 11,520 times **(d)** 1,152 times

9. A graph of height against age for a typical teenager shows: 1

(a) a fall **(b)** a rapid rise during the growth spurt
(c) no change **(d)** a straight line only

10. Which of the following shows a correct application of convection? 1

(a) sea breeze **(b)** spoon heating **(c)** cloth drying **(d)** sunlight on skin

11. A pendulum was timed for 10 oscillations: 18.2 s, 18.0 s, 18.4 s. The best estimate of T is: 1

(a) 18.2 s **(b)** 1.82 s **(c)** 1.8 s **(d)** 0.55 s

12. The reason that the human small intestine is long and coiled is: 1

(a) to provide a large surface area for absorption
(b) to make a noise
(c) to digest water
(d) to store food

13. **Assertion (A):** Adolescence is a period of rapid growth in height and body changes. 1
Reason (R): Hormones secreted during puberty control these changes.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** Heat always flows from a hotter body to a colder body. 1
Reason (R): Heat flows until the two bodies reach the same temperature.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The SI unit of time is the second. 1
Reason (R): One minute is equal to 60 seconds.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Explain why the endocrine system is called the 'chemical messenger system'. 2

17. Explain why heating a liquid from the top does not heat it quickly but from the bottom does. 2

18. Calculate the speed of a train in km/h if it crosses a 300 m bridge in 30 s and its length is 200 m. 2

19. Explain why a person feels warm after eating food rich in carbohydrates. 2

20. Why do boys have a higher proportion of muscle growth during adolescence? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain the effect of removing the pituitary gland in a child on growth and on other glands. 3

UNSTEPWISE

22. In an experiment, 100 mL of hot water at 80 °C is poured into three cups: metal, ceramic and thermocol. After 10 min the temperatures are 55 °C, 62 °C and 72 °C respectively. Arrange them and explain. 3

23. A train 120 m long crosses a pole in 12 s and a platform in 30 s. Find its speed and the length of the platform. 3

24. Explain the effects on the digestive system if the liver, pancreas and gall bladder are removed. 3

OR

A diabetic friend wants to join a picnic. What advice would you give about food and medicines? 3

25. Explain why two spoons, one of steel and one of plastic, in a cup of boiling water show different results. 3

OR

A bus leaves a town at 5 a.m. and runs at 40 km/h until 8 a.m. Find the distance covered. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Rahul, aged 13, notices that he is growing taller quickly, hair is appearing on his face, and his voice sometimes 'cracks' while speaking. He feels shy about these changes, but his teacher explains that they are normal. 5

(i) What is the stage of life in which these changes occur called?
(ii) Name the male sex hormone responsible for them.
(iii) Name the organ that secretes this hormone.
(iv) Explain why Rahul's voice becomes deeper.

27. A student plots the following data: time (s): 0, 5, 10, 15, 20; distance (m): 0, 10, 20, 20, 50. Describe the stages of motion, find the speed in each stage and the average speed. 5

OR

Design an investigation to find how the breathing rate changes with activity and present the data in a table with a conclusion. 5

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND